

Hunter Liles

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Education

McNeese State University

Expected May 2028

Computer Science and Computational Science

Technical Skills

Languages: C++, C, Rust, Python, Java, SQL, GLSL, HLSL

Graphics: OpenGL, Vulkan, GLFW

Tools: Git, Linux, LaTeX

Research Experience

Undergraduate Student Researcher

Present

McNeese State University - Engineering & Computer Science

Lake Charles, LA

- Work with another undergraduate researcher and a faculty supervisor to explore signed distance fields.

Selected Projects

Signed Distance Field Rendering Engine

C++, OpenGL, GLSL

- Built the C++ application and controls for a two-person SDF rendering project; my partner wrote most of the shaders.
- Wrote quaternion math for camera rotation and controls for ray travel distance and surface-hit tolerance; passed a scene ID from CPU to GPU to switch between SDF functions.
- Added shader hot reload by comparing saved files with a cached version, along with FPS and frame-time counters to help test changes.

Blossom - 2D Engine & Editor

C++, OpenGL

- Rewriting the engine in OpenGL with my own math library, including 2D/3D vectors and 4×4 matrices for camera, view, and projection calculations.
- Built a tile editor in the earlier prototype using an array of integer texture IDs to place and remove tiles, with binary files for loading and saving levels.
- Saved tile changes as they occurred and again when closing a level; added shader hot reload to the earlier prototype.

Terminal 3D Renderer

C, ANSI, POSIX termios

- Built a terminal point renderer in C with no third-party dependencies, using x/z and y/z perspective projection and custom transformations to terminal screen coordinates.
- Wrote a function to map a floating-point frame buffer to an ASCII gradient, currently setting point values manually; used termios for input settings and ANSI for display and color.

Additional Experience

Math & Science Tutor

Present

MAST Department, McNeese State University

Lake Charles, LA

- Help around five students a week with math through Calculus II, plus physics and computer science topics when needed.

Church IT

Present

Iowa First Baptist Church

Iowa, LA

- Chose and set up the church's replacement computer to fit its needs; continue to support that computer.